

Experiment No: 01

Date: 17/09/2026

Title: Student Utility Program using Switch Case and Control Statements

Objective / Aim

To design and implement a Student Utility Program in Java using switch statements, conditional statements, loops, and user input, demonstrating basic programming concepts such as menu-driven programming, decision-making, iteration, and mathematical calculations.

System Requirements

Hardware

- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage

Software

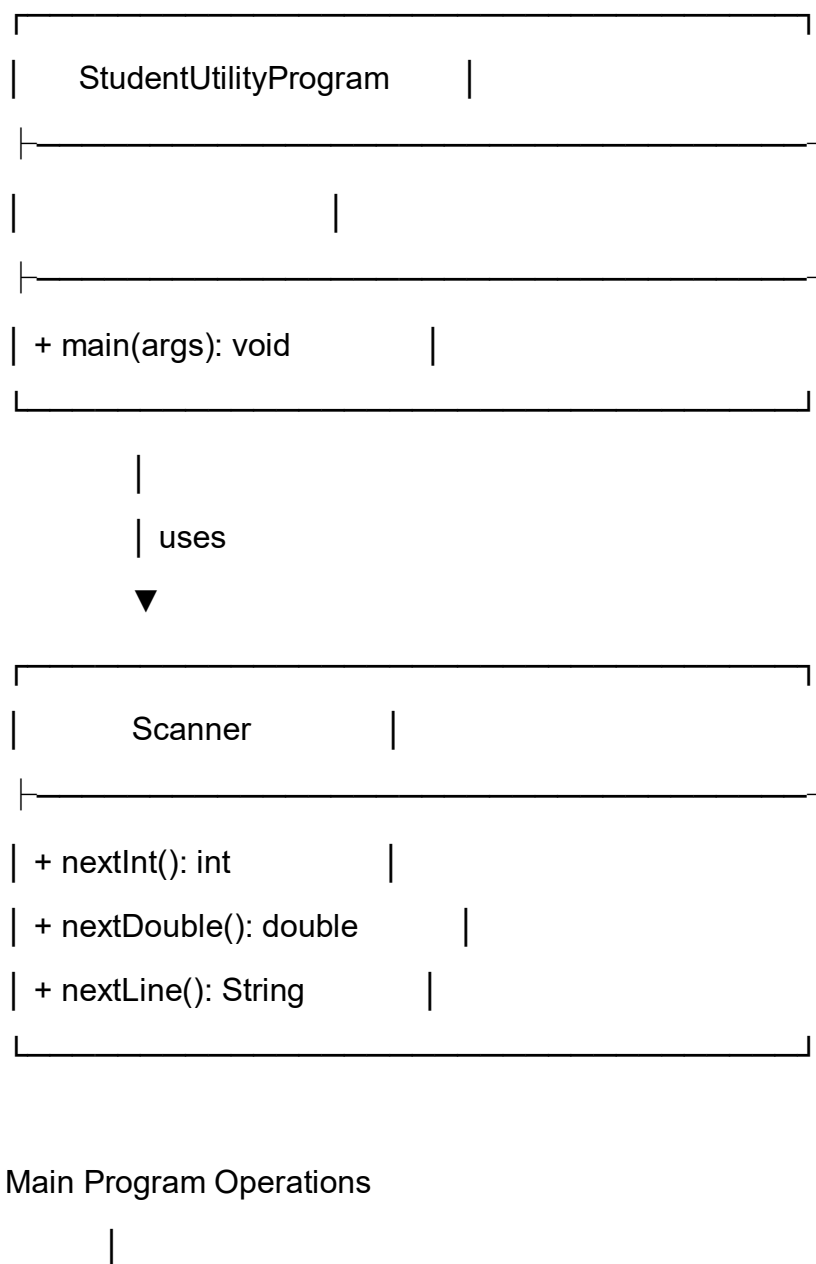
- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt / Terminal

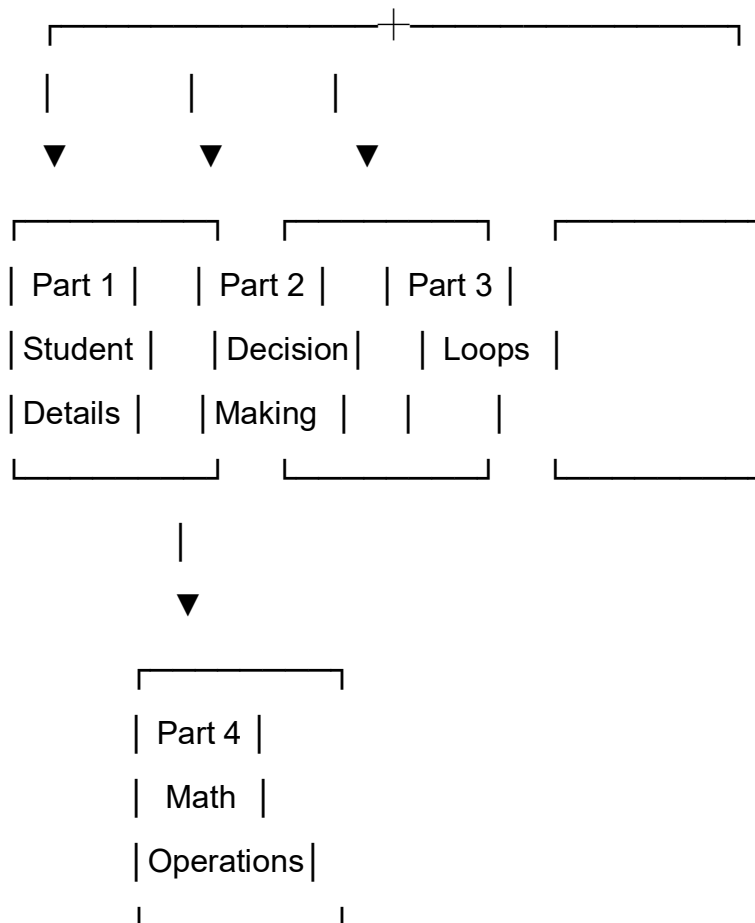
Algorithm / Logic Description

1. Create the main class named StudentUtilityProgram.
2. Create a "Scanner" object to accept input from the user.
3. Display the main menu containing four parts.
4. Accept the user's choice using a "switch" statement.
5. If the user selects Part 1, display a submenu for student information and marks.
6. Perform operations such as:
 - Display student information.
 - Accept student details and marks.
 - Calculate total and percentage.
 - Display complete student results.
7. If the user selects Part 2, display a submenu for decision-making operations.
8. Perform operations such as:
 - Check whether a number is even or odd.
 - Find the largest of three numbers.
 - Calculate grade based on percentage.
 - Display the day based on day number.
9. If the user selects Part 3, display a submenu for loop-based operations.
10. Perform operations such as:
 - Generate a multiplication table.
 - Display numbers from 1 to N.
 - Calculate the sum from 1 to N.
 - Generate the Fibonacci series.
11. If the user selects Part 4, display a submenu for mathematical operations.
12. Perform operations such as:
 - Calculate factorial.
 - Check whether a number is prime.
 - Find the maximum of two numbers.

- Calculate the area of a circle.
- 13. Display the appropriate result for the selected operation.
- 14. Close the Scanner object after execution.

UML Class Diagram





Relationship: The "StudentUtilityProgram" class uses the "Scanner" class for user input and uses switch-case, if-else, nested switch, and looping statements to perform different operations.

Source Code

```
package javacore;

import java.util.Scanner;

// Main class of the program
public class StudentUtilityProgram {
```

```

// Main method - program execution starts here
public static void main(String[] args) {

    // Creating Scanner object for user input
    Scanner sc = new Scanner(System.in);

    // Displaying main menu
    System.out.println("1. Part 1");
    System.out.println("2. Part 2");
    System.out.println("3. Part 3");
    System.out.println("4. Part 4");

    // Taking main menu choice from the user
    System.out.print("Enter your choice: ");
    int choice = sc.nextInt();

    // Main switch to select different parts
    switch (choice) {

        // ===== PART 1 =====

        case 1:

            // Displaying Part 1 sub-menu
            System.out.print("Enter case 1-4 ");
            int achoice = sc.nextInt();

            // Variables for student details and marks
            String name = "";

```

```
int rollno;

double marks1, marks2, marks3, total, percentage;

// Part 1 sub-menu
switch (achoice) {

    // Case 1: Display fixed student information
    case 1:
        System.out.println("Student Information");
        System.out.println("Name : Sameera");
        System.out.println("Roll No : 102");
        break;

    // Case 2: Accept student details and marks
    case 2:
        System.out.println("Enter Student Name: ");

        // Reading student name
        name = sc.nextLine();

        System.out.println("Enter Roll Number: ");
        rollno = sc.nextInt();

        // Reading English marks
        System.out.println("Enter English Marks: ");
        marks1 = sc.nextDouble();

        // Reading Maths marks
        System.out.print("Enter Maths Marks: ");
```

```
marks2 = sc.nextDouble();
```

```
// Reading Social marks
```

```
System.out.print("Enter Social Marks: ");
```

```
marks3 = sc.nextDouble();
```

```
break;
```

```
// Case 3: Calculate total and percentage
```

```
case 3:
```

```
System.out.print("Enter English Marks: ");
```

```
marks1 = sc.nextDouble();
```

```
System.out.print("Enter Maths Marks: ");
```

```
marks2 = sc.nextDouble();
```

```
System.out.print("Enter Social Marks: ");
```

```
marks3 = sc.nextDouble();
```

```
// Calculating total marks
```

```
total = marks1 + marks2 + marks3;
```

```
// Calculating percentage
```

```
percentage = (total / 300) * 100;
```

```
// Displaying result
```

```
System.out.println("Total = " + total);
```

```
System.out.println("Percentage = " + percentage);
```

```
break;
```

```
// Case 4: Accept details and display complete result
```

```
case 4:
```

```
    System.out.print("Enter Student Name: ");
```

```
    // Reading student name
```

```
    name = sc.nextLine();
```

```
    System.out.print("Enter Roll Number: ");
```

```
    rollno = sc.nextInt();
```

```
    // Reading English marks
```

```
    System.out.print("Enter English Marks: ");
```

```
    marks1 = sc.nextDouble();
```

```
    // Reading Maths marks
```

```
    System.out.print("Enter Maths Marks: ");
```

```
    marks2 = sc.nextDouble();
```

```
    // Reading Social marks
```

```
    System.out.print("Enter Social Marks: ");
```

```
    marks3 = sc.nextDouble();
```

```
    // Calculating total marks
```

```
    total = marks1 + marks2 + marks3;
```

```
    // Calculating percentage
```

```
    percentage = (total / 300) * 100;
```

```
    // Displaying complete result
```

```

        System.out.println("\n----- RESULT -----");
        System.out.println("Name : " + name);
        System.out.println("Roll No : " + rollno);
        System.out.println("Total : " + total);
        System.out.println("Percentage : " + percentage);
        break;

// Invalid Part 1 choice
default:
    System.out.println("Invalid Choice");
}

break;

// ===== PART 2 =====

case 2:

// Displaying Part 2 sub-menu
System.out.println("enter case 1-4");
int bchoice = sc.nextInt();

// Part 2 sub-menu
switch (bchoice) {

// Case 1: Check whether a number is even or odd
case 1:
    System.out.print("Enter a number: ");
    int n = sc.nextInt();

```

```
// Checking divisibility by 2
if (n % 2 == 0)
    System.out.println("Even Number");
else
    System.out.println("Odd Number");

break;
```

```
// Case 2: Find the largest of three numbers
```

```
case 2:
```

```
System.out.print("Enter first number: ");
int a = sc.nextInt();
```

```
System.out.print("Enter second number: ");
int b = sc.nextInt();
```

```
System.out.print("Enter third number: ");
int c = sc.nextInt();
```

```
// Comparing three numbers
```

```
if (a >= b && a >= c)
    System.out.println("Largest = " + a);
else if (b >= a && b >= c)
    System.out.println("Largest = " + b);
else
    System.out.println("Largest = " + c);
```

```
break;
```

```
// Case 3: Find grade based on percentage
```

```
case 3:
```

```
    System.out.print("Enter percentage: ");
```

```
    double per = sc.nextDouble();
```

```
// Checking percentage and assigning grade
```

```
if (per >= 90)
```

```
    System.out.println("Grade A");
```

```
else if (per >= 75)
```

```
    System.out.println("Grade B");
```

```
else if (per >= 60)
```

```
    System.out.println("Grade C");
```

```
else if (per >= 40)
```

```
    System.out.println("Grade D");
```

```
else
```

```
    System.out.println("Fail");
```

```
break;
```

```
// Case 4: Display day based on day number
```

```
case 4:
```

```
    System.out.print("Enter day number (1-7): ");
```

```
    int day = sc.nextInt();
```

```
// Nested switch for day selection
```

```
switch (day) {  
  
    case 1:  
        System.out.println("Monday");  
        break;  
  
    case 2:  
        System.out.println("Tuesday");  
        break;  
  
    case 3:  
        System.out.println("Wednesday");  
        break;  
  
    case 4:  
        System.out.println("Thursday");  
        break;  
  
    case 5:  
        System.out.println("Friday");  
        break;  
  
    case 6:  
        System.out.println("Saturday");  
        break;  
  
    case 7:  
        System.out.println("Sunday");  
        break;  
}
```

```
        // Invalid day number
        default:
            System.out.println("Invalid Day");
    }

    break;
}
```

```
// ===== PART 3 =====
```

```
case 3:
```

```
    // Displaying Part 3 sub-menu
    System.out.println("enter case 1-4");
    int cchoice = sc.nextInt();
```

```
    // Part 3 sub-menu
    switch (cchoice) {
```

```
        // Case 1: Display multiplication table
```

```
        case 1:
```

```
            System.out.print("Enter a number: ");
            int num = sc.nextInt();
```

```
            System.out.println("Multiplication Table:");
```

```
            // Loop from 1 to 10
```

```
            for (int i = 1; i <= 10; i++) {
```

```
        System.out.println(
            num + " x " + i + " = " + (num * i)
        );
    }
}
```

```
break;
```

```
// Case 2: Display numbers from 1 to N
```

```
case 2:
```

```
    System.out.print("Enter N: ");
```

```
    int n = sc.nextInt();
```

```
    System.out.println(
```

```
        "Numbers from 1 to " + n + ":"
```

```
    );
```

```
// Printing numbers from 1 to N
```

```
for (int i = 1; i <= n; i++) {
```

```
    System.out.print(i + " ");
```

```
}
```

```
break;
```

```
// Case 3: Calculate sum from 1 to N
```

```
case 3:
```

```
    System.out.print("Enter N: ");
```

```
    int m = sc.nextInt();
```

```
// Variable to store sum
int sum = 0;

// Adding numbers from 1 to N
for (int i = 1; i <= m; i++) {
    sum = sum + i;
}

// Displaying sum
System.out.println("Sum = " + sum);

break;
```

```
// Case 4: Generate Fibonacci series
case 4:
    System.out.print("Enter number of terms: ");
    int terms = sc.nextInt();
```

```
// Initial Fibonacci values
int a = 0, b = 1, c;
```

```
System.out.println("Fibonacci Series:");
```

```
// Generating Fibonacci series
for (int i = 1; i <= terms; i++) {
```

```
    // Display current term
```

```
System.out.print(a + " ");
```

```
// Calculate next term
```

```
c = a + b;
```

```
// Update Fibonacci values
```

```
a = b;
```

```
b = c;
```

```
}
```

```
break;
```

```
}
```

```
// ===== PART 4 =====
```

```
case 4:
```

```
// Displaying Part 4 sub-menu
```

```
System.out.println("enter case 1-4");
```

```
int dchoice = sc.nextInt();
```

```
// Part 4 sub-menu
```

```
switch (dchoice) {
```

```
// Case 1: Calculate factorial
```

```
case 1:
```

```
System.out.print("Enter a number: ");
```

```
int n = sc.nextInt();
```

```
// Initial factorial value
int fact = 1;

// Calculating factorial using loop
for (int i = 1; i <= n; i++) {
    fact = fact * i;
}

// Displaying factorial
System.out.println("Factorial = " + fact);

break;
```

```
// Case 2: Check whether a number is prime
case 2:
```

```
System.out.print("Enter a number: ");
int num = sc.nextInt();
```

```
// Variable to count factors
int count = 0;
```

```
// Checking all possible divisors
for (int i = 1; i <= num; i++) {
```

```
    if (num % i == 0) {
        count++;
    }
}
```

```
// A prime number has exactly two factors
if (count == 2)
    System.out.println("Prime Number");
else
    System.out.println("Not a Prime Number");

break;
```

```
// Case 3: Find maximum of two numbers
```

```
case 3:
```

```
System.out.print("Enter first number: ");
int a = sc.nextInt();
```

```
System.out.print("Enter second number: ");
int b = sc.nextInt();
```

```
// Comparing two numbers
```

```
if (a > b)
    System.out.println("Maximum = " + a);
else
    System.out.println("Maximum = " + b);

break;
```

```
// Case 4: Calculate area of a circle
```

```
case 4:
```

```
        System.out.print("Enter radius: ");
        double r = sc.nextDouble();

        // Calculating area of circle
        double area = 3.14 * r * r;

        // Displaying area
        System.out.println(
            "Area of Circle = " + area
        );

        break;

// Invalid Part 4 choice
default:
    System.out.println("Invalid Choice");
}

break;
}

// Closing Scanner object
sc.close();
}
}

---
```

Compilation & Execution Commands

Save the source code as:

StudentUtilityProgram.java

If you are using the "package javacore;" statement, save the file inside the appropriate "javacore" package/folder.

Compile the program:

```
javac StudentUtilityProgram.java
```

Run the program:

```
java StudentUtilityProgram
```

Output Screen

Example 1 – Student Information

1. Part 1

2. Part 2

3. Part 3

4. Part 4

Enter your choice: 1

Enter case 1-4 1

Student Information

Name : Sameera

Roll No : 102

Example 2 – Grade Calculation

1. Part 1

2. Part 2

3. Part 3

4. Part 4

Enter your choice: 2

enter case 1-4

3

Enter percentage: 85

Grade B

Example 3 – Multiplication Table

1. Part 1

2. Part 2

3. Part 3

4. Part 4

Enter your choice: 3

enter case 1-4

1

Enter a number: 5

Multiplication Table:

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

$$5 \times 3 = 15$$

$$5 \times 4 = 20$$

$$5 \times 5 = 25$$

$$5 \times 6 = 30$$

$$5 \times 7 = 35$$

$$5 \times 8 = 40$$

$$5 \times 9 = 45$$

$$5 \times 10 = 50$$

Example 4 – Factorial

1. Part 1

2. Part 2

3. Part 3

4. Part 4

Enter your choice: 4

enter case 1-4

1

Enter a number: 5

Factorial = 120

Result

The Student Utility Program was successfully implemented in Java using switch-case statements, if-else statements, nested switch, I